

Certified finite-block influences for the square-lattice Ising model

Cambyse Rouzé Daniel Stilck França

Numerical supplement, version 1.0.0

This note documents the reproducibility package for *Rapid mixing of Gibbs samplers via quantum Dobrushin–Shlosman conditions*. It records the finite-volume calculation and explicit constants used in the Ising example. References to general quantum-dynamical constructions and perturbation estimates refer to that companion manuscript.

1 Certification of the Ising block example

1.1 Finite-block influences and numerical certificates

On the square torus with L^2 sites, consider

$$H(g) = -J \sum_{\langle i,j \rangle} Z_i Z_j - g \sum_i X_i, \quad x = \beta J. \quad (1)$$

At $g = 0$, the exact single-site influence is $c_{ij} = \frac{1}{2} \tanh(2x)$, so the single-site certificate is

$$\kappa(x, 1) = 1 - 2 \tanh(2x). \quad (2)$$

It fails for $x \geq \frac{1}{2} \operatorname{arctanh}(1/2) \simeq 0.27465$. We exhibit positive block certificates at $x = 0.35$ and $x = 0.36$ by a finite computation, then apply the classical-reference perturbation theorem of the companion manuscript.

Take all N translates of an $\ell \times \ell$ square, each with rate $a_B = \ell^{-2}$, and assume $L \geq 2\ell + 2$. Every site belongs to exactly ℓ^2 blocks. Each block has 4ℓ distinct exterior neighbors. Write $c_p(x, \ell)$ for c_{Bj} when j meets position p along one side; rotations identify the four sides and reflection gives $c_p = c_{\ell+1-p}$.

Lemma 1.1 (Exact influence from ferromagnetic monotonicity). *Let $m_i^{\xi, \pm j}$ be the magnetization at $i \in B$ with the exterior spin j fixed to ± 1 and the other exterior spins fixed to ξ . For the ferromagnetic reference in (1),*

$$c_{Bj} = \frac{1}{2} \max_{\xi} \sum_{i \in B} (m_i^{\xi, +j} - m_i^{\xi, -j}), \quad (3)$$

where the maximum ranges over the other $4\ell - 1$ exterior neighbors.

Proof. Ferromagnetic monotonicity gives a coupling of the two conditional Gibbs laws with $\sigma \geq \sigma'$ coordinatewise; it follows, for example, by coupling their single-site heat baths with common uniform random numbers and passing to stationarity. In this coupling the Hamming distance is $\frac{1}{2} \sum_{i \in B} (\sigma_i - \sigma'_i)$, so its expectation gives the upper bound in (3). The function $f(\sigma) = \frac{1}{2} \sum_{i \in B} \sigma_i$ is 1-Lipschitz for Hamming distance, giving the reverse bound by Kantorovich duality. Only exterior neighbors enter the conditional Hamiltonian. $\square$

For a fixed site j , each side position occurs in four blocks that have j as an exterior neighbor. Thus the exact classical and lifted margins coincide and equal

$$\kappa(x, \ell) = 1 - \frac{4}{\ell^2} \sum_{p=1}^{\ell} c_p(x, \ell). \quad (4)$$

We evaluate (3) by a column transfer matrix with 2^ℓ states, tracking the partition function and the expected number of positive spins. The search exhausts all boundary configurations, with global spin flip removing one binary choice. Positive-arithmetic error bounds enclose every computed c_p within 10^{-9} ; Subsection 1.2 gives the algorithm and certification details.

ℓ	$x = 0.30$	$x = 0.35$	$x = 0.40$	$x = 0.42$
1	-0.074099	-0.208736	-0.328074	-0.371618
2	0.035426	-0.186278	-0.417502	-0.511628
3	0.123728	-0.170668	-0.520613	-0.675481
4	0.229515	-0.097434	-0.533118	-0.741202
5	0.317415	-0.026701	-0.536402	-0.798591
6	0.391737	0.043216	-0.525389	-0.837924
First ℓ with $\kappa > 0$	2	6	> 6	> 6
First ℓ with $\kappa \geq 1/2$	> 6	> 6	> 6	> 6

Table 1: Square-block margins $\kappa(x, \ell)$, rounded to six decimal places. Each unrounded value has certified error below 4.1×10^{-9} . The last two rows refer to the search $1 \leq \ell \leq 6$; > 6 means that no tested size passed the stated test.

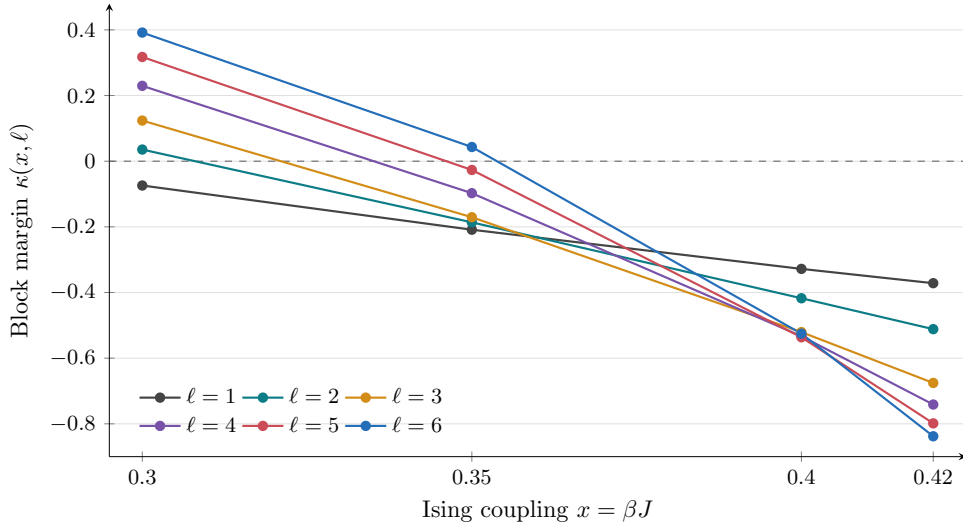


Figure 1: The certified margins in Table 1. Positive values give block certificates. Lines join the computed points as visual guides.

In particular, $(x, \ell) = (0.35, 6)$ has margin greater than 0.0432. One further size gives $\kappa(0.36, 7) > 0.0125$, whereas (2) is negative at both temperatures. For either pair choose the displayed lower bound as κ . The finite block certificate is the hypothesis used in this application. Strong mixing for squares [1] and the finite-volume DS tests approaching criticality [2] provide the broader classical context.

1.2 Exhaustive boundary search and arithmetic bounds

We describe the finite computation underlying Table 1 and the two positive certificates in the Ising example of the companion manuscript. The code, certified input weights, full coefficient tables, and maximizing boundary configurations are supplied in `numerical/ising_blocks/`.

For a column $s \in \{-1, +1\}^\ell$, let $n_+(s)$ count its positive spins. If t_c, b_c are the top and bottom boundary spins at column c , define

$$K(s, s') = e^{x \sum_{i=1}^{\ell} s_i s'_i}, \quad V_c(s) = \exp \left[x \left(\sum_{i=1}^{\ell-1} s_i s_{i+1} + t_c s_1 + b_c s_\ell \right) \right].$$

Keep the entire left boundary a as a matrix index. Initialize $Z_1(a, s) = K(a, s)V_1(s)$ and $U_1(a, s) = n_+(s)Z_1(a, s)$, and iterate

$$\begin{aligned} Z_{c+1}(a, s') &= V_{c+1}(s') \sum_s K(s, s') Z_c(a, s), \\ U_{c+1}(a, s') &= V_{c+1}(s') \sum_s K(s, s') [U_c(a, s) + n_+(s') Z_c(a, s)]. \end{aligned}$$

Finally multiply each matrix by K to incorporate every right boundary b . The ratio $U(a, b)/Z(a, b)$ is the expected total number of positive spins. Differences between left boundary rows differing at position p therefore give exactly the quantities maximized in (3). The matrix K is a tensor product of ℓ two-by-two matrices, so each transfer is evaluated by successive two-component transforms.

Global spin flip preserves each response difference, allowing the first top boundary spin to be fixed to $+1$. We enumerate all $2^{2\ell-1}$ remaining top/bottom patterns, while the two matrix indices cover all left/right patterns. For each p , this checks all $2^{4\ell-2}$ representatives of the other exterior spins. Reflection is used to check $c_p = c_{\ell+1-p}$, rather than to restrict the maximizing boundary. At $(x, \ell) = (0.35, 6)$ the three inequivalent coefficients, each with error below 10^{-9} , are

$$c_1 = 1.10740048984, \quad c_2 = 1.52878638950, \quad c_3 = 1.66933990243.$$

They give $\kappa(0.35, 6) > 0.0432162700$. The additional search at $(0.36, 7)$ gives $\kappa(0.36, 7) > 0.0125511558$.

Here is an arithmetic certificate for all entries used in the paper. The input values x are exact terminating decimals. The weights e^{kx} , $|k| \leq 9$, are computed with a correctly rounded 100-digit decimal exponential; an enclosure of width 2×10^{-98} certifies that each binary64 weight has relative error at most 2^{-52} . The search uses round-to-nearest binary64 arithmetic without reassociation. Every entry of Z and U is assembled from nonnegative sums and products. For $\ell \leq 8$, the number of unit-roundoff factors along any operation path is at most $4\ell^2 + 5\ell + 5 \leq 301$. With $\epsilon_{\text{fp}} = 2^{-53}$, the conservative bound

$$\gamma = \frac{4096\epsilon_{\text{fp}}}{1 - 4096\epsilon_{\text{fp}}} < 4.55 \times 10^{-13}$$

therefore bounds the relative error of every positive Z or U ; a zero U is represented exactly. All positive intermediates lie between e^{-61} and $64 \cdot 2^{64} e^{61}$ for the parameters used, so there is neither overflow nor underflow.

The ratio U/Z lies in $[0, \ell^2]$. Its absolute error is at most

$$\ell^2 \left[\frac{1 + \gamma}{1 - \gamma} (1 + \epsilon_{\text{fp}}) - 1 \right].$$

Subtracting two such ratios gives error below 1.3×10^{-10} for $\ell \leq 8$. Taking a maximum preserves this absolute-error bound, so we enclose each c_p by the computed maximum plus or

minus 10^{-9} , with endpoints rounded outward. The corresponding margin has error at most $4 \times 10^{-9}/\ell + 10^{-13}$, including the final summation. An independent enumeration of all interior and exterior spin configurations for $\ell = 1, 2, 3$ agrees with the transfer computation to within 2×10^{-13} . It also verifies (2), zero influence at $x = 0$, and the small-coupling limit.

1.3 Explicit shell constants for the square-lattice example

We give explicit perturbation windows for the block certificates in Subsection 1.1. Rescale the Hamiltonian by β , so that the inverse temperature is one, the edge norm is $x = \beta J$, and the on-site perturbation norm is $u = \beta|g| \leq u_0 \leq 1$. Let B be an $\ell \times \ell$ square on a torus with $L \geq \ell + 2$, $R_B = B^{\leq 1}$, and $R_{B,r} = B^{\leq r+1}$. Put

$$b = \ell^2, \quad r_0 = \ell^2 + 4\ell, \quad \chi = r_0/b, \quad w = 4\ell x, \quad T = e^{4\ell x} \log \frac{32r_0\chi}{\kappa}, \quad (5)$$

where κ is the chosen certified lower bound on the block margin. The on-site perturbation changes no crossing interaction, so the local perturbation bound of the companion manuscript uses $E = e^w$. The corresponding infinite-lattice neighborhood has cardinality $\ell^2 + 4\ell(r+1) + 2r(r+1) \leq r_0(r+1)^2$. Passing to the torus can only reduce this cardinality, and translation invariance gives weighted overlap $|R_{B,r}|/b$. Thus the polynomial-growth and weighted-overlap hypotheses of the companion manuscript holds with $D = 2$, $V = r_0$, and $\Xi = \chi$.

An explicit real-time restriction bound is

$$\|\tau_t^K(A) - \tau_t^{K'}(A)\| \leq 8|S|\|A\| \min\{1, (e^{v|t|} - 1)e^{-q}\}, \quad v = 32ex, \quad (6)$$

when A is supported on S and the smaller restriction contains $S^{\leq q}$. To verify the constants, remove all on-site terms in the interaction picture. The remaining time-dependent edge interactions still have norm at most x . More explicitly, for a fixed observable F define $C_F(S, t) = \sup_{A \in \mathcal{B}(S)} \|[\tau_t^{K'}(A), F]\|/\|A\|$. The usual interaction-picture commutator recurrence is bounded by

$$C_F(S, t) \leq C_F(S, 0) + 2x \sum_{e: e \cap S \neq \emptyset} \int_0^{|t|} C_F(e, s) ds, \quad C_F(S, 0) \leq 2\|F\| \mathbf{1}_{S \cap \text{supp } F \neq \emptyset}.$$

Enlarging the edge sum to all edges meeting S only increases it. Its iteration is over consecutive overlapping edges, rather than successively growing operator supports: there are at most $4|S|$ choices of the first edge and at most seven choices of each subsequent edge overlapping its predecessor. Apply this bound with F each edge term crossing the restriction boundary in Duhamel's formula. For n interior edges, summing also over the final boundary edge costs at most $4|S|7^n$, and only $n \geq q$ can reach that boundary. The norm of the final edge is at most x and its commutator contributes a factor two. Integrating $8x|S|\|A\| \sum_{n \geq q} (14xs)^n/n!$ over $0 \leq s \leq |t|$ gives

$$\|\tau_t^K(A) - \tau_t^{K'}(A)\| \leq \frac{4}{7}|S|\|A\| \sum_{n \geq q+1} \frac{(14x|t|)^n}{n!} \leq \frac{4}{7e}|S|\|A\|e^{-q}(e^{14ex|t|} - 1).$$

Indeed, a chain reaching the boundary has at least q interior edges before the final Duhamel edge. The trivial bound $2\|A\|$ then proves (6). This argument also applies to weakened crossing edges in the QBP interpolation and to a passive reference.

Set

$$\bar{v} = \max\{1, 32ex\}, \quad \mu = \frac{1}{2\bar{v}}, \quad C_{\text{tr}} = 36 + 4\bar{v}. \quad (7)$$

The kernel representations imply, separately for $\mathcal{Q} = \mathcal{F}$ and $\mathcal{Q} = \mathcal{G}$,

$$\|\mathcal{Q}_K(A) - \mathcal{Q}_{K'}(A)\| \leq C_{\text{tr}}|S|\|A\|e^{-\mu q}. \quad (8)$$

For completeness, split the time integrals at $t_q = (q+1)/(2\bar{v})$. The positive kernel satisfies $\int_{|t|>t_q} p_1(t) dt \leq e^{-2\pi t_q}$. The short-time part of its integral is bounded by $8e^{1/2}|S|\|A\|e^{-q/2}$, and its long-time part by $2\|A\|e^{-\pi(q+1)/\bar{v}}$. For the odd kernel, the short-time part is bounded by

$$\frac{8}{\pi}|S|\|A\|e^{-q} \int_0^{(q+1)/2} \frac{e^z - 1}{z} dz \leq \frac{8e^{1/2}}{\pi}|S|\|A\|e^{-q/2}.$$

The long-time part is at most

$$\frac{2}{\pi}\|A\| \log \coth(\pi t_q) \leq \frac{4}{\pi}(1 + \bar{v}/\pi)\|A\|e^{-\pi(q+1)/\bar{v}}.$$

Here we used $\log \coth z \leq 2e^{-2z}/(1 - e^{-2z})$ and $(1 - e^{-a})^{-1} \leq 1 + a^{-1}$. The choices in (7) dominate both bounds, including $q = 0$.

Define the following finite constants:

$$\begin{aligned} E &= e^w, & P &= 2^{r_0}, & a_0 &= r_0/2, & J_0 &= 1 + a_0 u_0, \\ K_Q &= EC_{\text{tr}} r_0 w, & L_Q &= \max\{E, e^\mu K_Q\}, \\ K_P &= 4PJ_0 C_{\text{tr}} r_0, & L_P &= \max\{Pa_0(2 + a_0 u_0)u_0, e^\mu K_P\}, \\ C_s &= (8r_0 + 34)(K_Q + K_P) + 2C_{\text{tr}} r_0(L_Q + L_P). \end{aligned} \tag{9}$$

Then the augmented physical generators satisfy

$$\|\mathcal{A}_{B, R_B, r+1}(H) - \mathcal{A}_{B, R_B, r}(H)\|_\diamond \leq C_s(r+1)^3 e^{-\mu r}. \tag{10}$$

To check this propagation, the dressing and its inverse have norms at most $e^{w/4}$. Equation (8) and variation of constants bound each successive dressing difference by $e^{w/4}(C_{\text{tr}} r_0 w/4)e^{-\mu r}$. Telescoping its four conjugating factors bounds the successive QBP jump-map and loss differences by $K_Q e^{-\mu r}$. For pinching, each jump has norm at most J_0 and successive difference at most $2C_{\text{tr}} r_0 e^{-\mu r}$. Summing over the P projectors therefore bounds the jump-map and loss differences by $K_P e^{-\mu r}$.

For either part of the loss, write $A_m = S_m - S_{m-1}$ for $m \geq 1$, with $A_0 = S_0$ for QBP and $A_0 = S_0 - I$ for pinching. The corresponding constants above give $\|A_m\| \leq L e^{-\mu m}$ and $\|S_{r+1} - S_r\| \leq K e^{-\mu r}$. The coherent shell is

$$\mathcal{G}_{H_{R_B, r+1}}(A_{r+1}) + \sum_{m=0}^r [\mathcal{G}_{H_{R_B, r+1}} - \mathcal{G}_{H_{R_B, r}}](A_m).$$

The smaller region contains the $(r-m)$ -neighborhood of the support $R_{B, m}$; thus the distance in the m th term is exactly $q = r - m$, with no shell offset. The multiplier bound in the coherent-multiplier lemma of the companion manuscript and (8) give respectively

$$(r_0 + 4)K(r+2)^2 e^{-\mu r}, \quad C_{\text{tr}} r_0 L \sum_{m=0}^r (m+1)^2 e^{-\mu r}.$$

Their sum is at most $[4(r_0+4)K + C_{\text{tr}} r_0 L](r+1)^3 e^{-\mu r}$. Adding the jump-map and anticommutator differences, and twice the coherent difference, proves (10) with (9).

The infinite tail needed in the perturbation-threshold formula of the companion manuscript is bounded without numerical truncation. For $R+2 \geq 14/\mu$,

$$\sum_{r \geq R} (r+2)^7 e^{-\mu r} \leq \frac{(R+2)^7 e^{-\mu R}}{1 - e^{-\mu/2}}, \tag{11}$$

because the ratio of successive summands is at most $e^{-\mu/2}$. Use $V = r_0$, $\Xi = \chi$, C_{loc} from the local-continuity formula of the companion manuscript with $E = e^w$, and the constants above in the aggregate perturbation-error formula of the companion manuscript. Choose R_* so the majorant multiplied by B_0 is at most $\kappa/8$, and choose $u_* \leq \min\{u_0, \kappa/[8A(1 + R_*)^8]\}$. The supplied script `numerical/ising_blocks/perturbation_window.py` performs this calculation using outward-rounded decimal arithmetic. With $u_0 = 1$, it gives the sufficient choices

x	ℓ	κ	T (approximately)	R_*	u_*
0.35	6	0.0432	4.9864×10^4	9481	10^{-68}
0.36	7	0.0125	3.0169×10^5	10800	10^{-76}

The margins in this table are certified lower bounds supplied by the separate finite-volume computation. These small sufficient windows reflect the conservative continuity and shell constants.

Remark 1.1 (The boundary cost of two-dimensional blocks). For these squares, $w = 4\ell x$ grows with the perimeter and the guaranteed interior erasure rate is $e^{-4\ell x}$. Thus larger blocks improve the classical certificate at the cost of a longer internal time and a smaller sufficient perturbation window. In one dimension the boundary has bounded size independently of the block length. The windows here establish existence; we expect the admissible range of $|\beta g|$ at these fixed temperatures to be of order one.

References

- [1] F. Martinelli, E. Olivieri, and R. H. Schonmann. For 2-D lattice spin systems weak mixing implies strong mixing. *Communications in Mathematical Physics* **165**, 33–47 (1994). [doi:10.1007/BF02099735](https://doi.org/10.1007/BF02099735).
- [2] S. B. Shlosman. The critical temperature $T_{cr}(\text{Ising})$ is DS-computable (2025). [arXiv:2505.24750v2](https://arxiv.org/abs/2505.24750v2).